Colon & Colorectal Cancer — Biomarker Testing Intelligence

Alert Window: Last 30 Days · Generated: 2026-04-20 07:30:02

Situation Summary

Stage IV: 3 of 22
Biomarker Count: 14 entries
Top Targets: ctDNA (5) · MSI/MMR (5) · TP53 (1)
Breakthrough: 7 entries scored ≥ 4

Research & Guidelines Timeline

Biomarker testing entries ranked by significance score. Higher scores indicate greater clinical relevance. Always discuss with your oncologist.

MEDIUM Stage IV BROAD SCOPE VERIFIED GENOMIC PROFILING PUBLISHED MAR 28, 2026 9

The Evolving Role for Repeat Molecular Testing in Metastatic Colorectal Cancer

Next-generation sequencing (NGS) is integral in managing metastatic colorectal cancer (mCRC), enhancing outcomes via targeted therapies and immune checkpoint inhibitors. NCCN and ASCO guidelines recommend initial testing for RAS, BRAF, HER2, and microsatellite instability (MSI)/mismatch repair (MMR) status to guide treatment decisions. The study underscores the importance of repeat molecular testing to adapt treatment strategies as tumor biology evolves.

pubmed.ncbi.nlm.nih.gov

HIGH All Stages ctDNA VERIFIED LIQUID BIOPSY PUBLISHED APR 03, 2026 7

Liquid biopsies using circulating tumor DNA for surveillance of gastrointestinal cancers in Hispanics: first real-world data report

Integration of circulating tumor DNA (ctDNA) testing in a community oncology setting for gastrointestinal cancers, primarily colorectal cancer (CRC), demonstrates potential for effective disease monitoring. This study presents real-world data supporting the adoption of ctDNA testing in routine clinical practice.

pubmed.ncbi.nlm.nih.gov

HIGH All Stages BROAD SCOPE VERIFIED LIQUID BIOPSY PUBLISHED MAR 30, 2026 6

Utility of Urinary Extracellular Vesicles for Colorectal Cancer: Comparison of DNA Quality and MRD Detection

Urinary extracellular vesicles (EVs) were assessed as a liquid biopsy source for colorectal cancer (CRC) in two cohorts (97 for validation, 63 for survival analysis). The study compared urinary EV-DNA, urinary cell-free tumor DNA (ctDNA), plasma EVs, and plasma ctDNA. Urinary EVs demonstrated larger DNA fragments and showed potential for minimal residual disease (MRD) detection, indicating their utility in CRC monitoring.

pubmed.ncbi.nlm.nih.gov

HIGH Early Detection ctDNA VERIFIED LIQUID BIOPSY PUBLISHED MAR 25, 2026 4

Blood-based circulating tumour DNA (ctDNA) tests for colorectal cancer screening: Systematic review and meta-analysis of diagnostic accuracy

ctDNA-based blood testing shows high specificity and clinically relevant accuracy for invasive colorectal cancer (CRC) detection, though it has limited efficacy for identifying precancerous lesions. The findings support the complementary role of ctDNA in population screening, potentially enhancing participation and access to CRC prevention. However, further large-scale studies are necessary to validate its clinical and economic viability.

pubmed.ncbi.nlm.nih.gov

HIGH | All Stages | ctDNA | VERIFIED | LIQUID BIOPSY | PUBLISHED APR 17, 2026 | 3

How Survivors of Colorectal Cancer Experience ctDNA Testing and Fear of Recurrence

In a cohort of colorectal cancer survivors, ctDNA-informed adjuvant chemotherapy (ACT) decisions did not increase fear of cancer recurrence (FCR) or cause long-term psychosocial harm, on average 5 years post-trial entry. The study highlights the importance of clear communication strategies as ctDNA testing becomes routine, ensuring patient understanding and trust in clinician-patient decision-making.

pubmed.ncbi.nlm.nih.gov

MEDIUM | All Stages | BROAD SCOPE | VERIFIED | RESEARCH | PUBLISHED APR 14, 2026 | 3

Development and validation of a multicenter nomogram predicting the risk of liver metastasis after curative resection of colorectal cancer

A nomogram integrating clinicopathological features with KRAS, BRAF, and MSI status was developed and validated across multiple centers. This model demonstrated improved predictive accuracy and generalizability over conventional staging systems, aiding in the identification of high-risk colorectal cancer patients for tailored postoperative surveillance strategies to enhance long-term survival outcomes.

pubmed.ncbi.nlm.nih.gov

MEDIUM | Stage I | BRAF | VERIFIED | RESEARCH | PUBLISHED APR 14, 2026 | 3

Increased Risk of Central Mesocolic Lymph Node Metastases in BRAF-Mutated Stage I-III Colon Cancer

BRAF mutation status was assessed in a prospective cohort of 97 stage I-III colon cancer patients from the T-REX trial. The study found that BRAF-mutated patients had an increased risk of central mesocolic lymph node metastases. This suggests that molecular profiling for BRAF mutations may guide decisions regarding the extent of lymphadenectomy, potentially improving surgical outcomes in this subset of patients.

pubmed.ncbi.nlm.nih.gov

MEDIUM | All Stages | MSI/MMR | VERIFIED | RESEARCH | PUBLISHED APR 14, 2026 | 3

The Keynote-177 Randomized Controlled Trial of First-line PD-1 Blockade for Microsatellite-High or Mismatch Repair-Deficient Advanced Colorectal Cancer: Potential Consequences of the Cross-Over Trial Design and Related Issues

The Keynote-177 trial examines first-line PD-1 blockade in patients with microsatellite-high or mismatch repair-deficient advanced colorectal cancer. The study highlights the implications of crossover trial design on treatment access and efficacy assessment. It underscores the challenges posed by high drug costs and the variability in health technology assessments, which may affect global access to effective therapies.

pubmed.ncbi.nlm.nih.gov

HIGH | Early Detection | ctDNA | VERIFIED | LIQUID BIOPSY | PUBLISHED APR 01, 2026 | 3

Circulating Tumor Cells as the Liquid Biopsy Foray into Noninvasive Colorectal Cancer Screening

Circulating tumor cells (CTCs) are being explored as a noninvasive biomarker for colorectal cancer screening. The study by Nguyen et al. assesses the feasibility of CTCs in comparison to traditional stool-based tests like the fecal immunochemical test (FIT). The findings suggest that CTCs may offer a complementary approach to existing screening methods, potentially enhancing early detection strategies in clinical practice.

pubmed.ncbi.nlm.nih.gov

MEDIUM All Stages MSI/MMR VERIFIED RESEARCH PUBLISHED APR 18, 2026 2

MSIanalyzer: Targeted Nanopore Sequencing Enables Single Nucleotide Resolution Analysis of Microsatellite Instability Diversity

MSIanalyzer utilizes targeted Oxford Nanopore sequencing to profile microsatellite instability (MSI) at single-nucleotide resolution, including base-level annotation of non-repeat interruptions. The study characterizes repeat-unit distributions in mismatch repair (MMR)-deficient and MMR-proficient cell lines and primary human tissues. This method enhances the understanding of MSI diversity, potentially informing therapeutic strategies in colorectal cancer.

pubmed.ncbi.nlm.nih.gov

MEDIUM Early Detection BROAD SCOPE VERIFIED EARLY DETECTION PUBLISHED APR 15, 2026 2

Immunohistochemical Expression of MLH1 and MSH2 in Colorectal Carcinoma and Its Correlation With Clinicopathological Parameters

MLH1 and MSH2 expression was assessed in colorectal carcinoma using immunohistochemistry to evaluate their correlation with clinicopathological parameters. The study highlights the significance of MMR deficiency in predicting outcomes, screening for Lynch syndrome, and guiding treatment decisions, particularly regarding immune checkpoint inhibitors.

pubmed.ncbi.nlm.nih.gov

MEDIUM All Stages Genomic Profiling VERIFIED RESEARCH PUBLISHED APR 13, 2026 1

Multi-layer molecular profiling defines an immune-active colorectal cancer subtype with therapeutic relevance

The study identifies three colorectal cancer subtypes (CS1-CS3) through a multi-omics approach integrating transcriptomic, epigenetic, and mutational profiles. The CS3 subtype is characterized by genomic instability and significant immune activation, suggesting its potential as a target for immunotherapy. This stratification may enhance patient selection for immune-based treatments in colorectal cancer.

pubmed.ncbi.nlm.nih.gov

MEDIUM Early Detection BROAD SCOPE VERIFIED GENOMIC PROFILING PUBLISHED MAR 27, 2026 1

Biomarkers in Colorectal Cancer: Clinically Relevant Diagnostic and Prognostic Molecular Features, and the Future of Precision Medicine

Colorectal cancer (CRC) remains a significant public health issue, with over 153,000 new cases and 53,000 deaths annually in the U.S. The study highlights the importance of understanding molecular and genetic biomarkers for CRC, which are critical for diagnosis, prognosis, and treatment planning. Specific gene mutations are emphasized as key factors in the management of CRC, underscoring the need for precision medicine approaches in clinical practice.

pubmed.ncbi.nlm.nih.gov

MEDIUM All Stages MSI/MMR VERIFIED RESEARCH PUBLISHED MAR 26, 2026 1

A PMS2-deficient pediatric high-grade glioma with PI3K-pathway mutations and adjacent developmental venous anomaly suggestive of CMMRD

PMS2 deficiency was identified in a pediatric high-grade glioma, associated with PI3K-pathway mutations and a developmental venous anomaly, indicative of constitutional mismatch repair deficiency (CMMRD). Routine mismatch repair immunohistochemistry is critical for diagnosis, genetic counseling, and potential therapeutic strategies. Integrated histological and genomic evaluation is essential for accurate diagnosis in this context.

pubmed.ncbi.nlm.nih.gov

MEDIUM All Stages MSI/MMR VERIFIED RESEARCH PUBLISHED MAR 24, 2026 1

dMMR prediction from colorectal cancer histopathology: Leveraging non-tumor and low-magnification regions

dMMR status in colorectal cancer can be predicted from histopathological whole slide images (WSIs) using machine learning (MIL-based models). The study shows moderate generalizability across independent cohorts and highlights that both tumor and non-tumorous tissue regions contain predictive signals, indicating that microenvironmental features may influence dMMR-associated histological patterns.

pubmed.ncbi.nlm.nih.gov

MEDIUM All Stages EGFR VERIFIED RESEARCH PUBLISHED MAR 23, 2026 1

Circulating Butyrate Attenuates Cetuximab Efficacy in Colorectal Cancer Through EGFR and AMPK-Wip1 Signaling

Butyrate may serve as a predictive biomarker for treatment response in KRAS wild-type colorectal cancer, as it reduces cetuximab efficacy via EGFR and AMPK-Wip1 signaling pathways.

pubmed.ncbi.nlm.nih.gov

Appendix A — Patient-Friendly Summary

Plain-language overview for patients, caregivers, and family members

What's happening

Biomarker testing — also called molecular profiling or genomic testing — analyzes your tumor's DNA to identify specific mutations. The results determine which FDA-approved treatments may be available to you. The entries below represent the highest-scoring study per biomarker target drawn from the full research archive — not limited to the current report window. Some entries may predate the timeline above; they are included because they remain the most clinically significant reference available for that biomarker.

ctDNA
Circulating Tumor DNA as a Biomarker for Recurrence After Curative Resection of Colorectal Cancer Liver Metastases: A Systematic Review and Meta-Analysis

Circulating tumor DNA (ctDNA) has been assessed as a biomarker for minimal residual disease detection in colorectal liver metastases (CRLM) post-curative resection. A systematic review and meta-analysis were conducted to quantify its prognostic value, addressing the high recurrence rates of 50-70% in this patient population. The findings suggest that ctDNA may serve as a significant indicator for recurrence risk in resected CRLM, aiding in post-operative monitoring strategies.

TP53
Clinical impact of TP53 mutation status on survival outcomes in metastatic colorectal cancer

TP53 mutations serve as an independent prognostic marker for prolonged progression-free survival in metastatic colorectal cancer. While overall survival differences were not statistically significant, the findings suggest the need for further validation in prospective studies to assess TP53's role in therapeutic stratification.

BRAF
Detection of KRAS, NRAS and BRAF Mutations in Liquid Biopsy from Patients with Colorectal Cancer

ddPCR analysis of circulating tumor DNA (ctDNA) in colorectal cancer patients reveals that it can provide complementary information for molecular diagnosis. The study highlights the utility of liquid biopsy in detecting KRAS, NRAS, and BRAF mutations, enhancing diagnostic accuracy in this patient population.

NGS
Exploratory biomarker findings from regorafenib plus toripalimab in patients with refractory metastatic colorectal cancer (REGOTORI study)

In the REGOTORI study, a multi-omics analysis identified several biomarkers associated with regorafenib plus toripalimab in refractory metastatic colorectal cancer. Key findings include ctDNA maxAF, HAF-bTMB, bITH, HAF-bITH, and mutational status of SMAD4 or PIK3CA, along with TME markers, which may serve as predictive or prognostic indicators for treatment response.

MSI/MMR
Prognostic Impact of Blood Tumor Mutational Burden in pMMR/MSS Metastatic Colorectal Cancer Assessed by FoundationOne( ) Liquid CDx

Blood tumor mutational burden (bTMB) was assessed using FoundationOne Liquid CDx in a monocentric, real-world study of patients with metastatic colorectal cancer (mCRC). The study aimed to clarify the prognostic significance of bTMB in an unselected cohort. Key findings regarding the clinical implications of bTMB in this context were not detailed in the abstract. Further results are necessary to determine its utility in prognostication for mCRC.

RAS
Effect of sex differences on the emergence of ctDNA RAS mutations in RAS wild-type colorectal cancer

In a single-center retrospective study of 43 patients with microsatellite stable RAS and BRAF wild-type metastatic colorectal cancer, the emergence of RAS mutations during chemotherapy was analyzed using a ctDNA assay. The study aimed to identify clinicopathological factors associated with this emergence, highlighting the potential impact of sex differences on RAS mutation development in RAS wild-type mCRC.

Genomic Profiling
Integrated Molecular Profiling of Colorectal Cancer by Tumor Location: Evidence from a Real-World Cohort with Primary and Metastatic Samples

Colorectal cancer (CRC) tumor location correlates with distinct molecular profiles. Right-sided tumors exhibit dMMR, MSI-H, and elevated TMB, while left-sided tumors show increased ERBB2 alterations and class 5 APC mutations. These findings underscore the necessity of incorporating tumor location into personalized molecular diagnostics and treatment strategies for CRC patients.

HER2

Targeting HER2 in Metastatic Colorectal Cancer: Current Therapies, Biomarker Refinement, and Emerging Strategies

HER2 overexpression/amplification occurs in 3-5% of metastatic colorectal cancers (mCRCs) and is a therapeutically actionable biomarker. The review discusses established and emerging HER2-targeted therapies, highlighting their incorporation into treatment guidelines based on increasing clinical evidence.

KRAS

Next-Generation Sequencing-Based Detection of KRAS G12D Variants in Colorectal Cancer: A Retrospective Cohort Study

KRAS G12D variants were assessed in a retrospective cohort of 73 colorectal cancer patients using next-generation sequencing. The study highlights the prevalence of KRAS variants, emphasizing the G12D variant's role as a predictor of resistance to anti-EGFR therapy. This finding underscores the importance of KRAS testing for guiding targeted treatment strategies in CRC.

EGFR

Circulating Butyrate Attenuates Cetuximab Efficacy in Colorectal Cancer Through EGFR and AMPK-Wip1 Signaling

Butyrate may serve as a predictive biomarker for treatment response in KRAS wild-type colorectal cancer, as it reduces cetuximab efficacy via EGFR and AMPK-Wip1 signaling pathways.

miRNA

Diagnostic Performance of Circulating miRNA-92a and Related Circulating miRNA Panels for Colorectal Cancer: An Updated Systematic Review and Meta-Analysis

Circulating miRNA-92a shows superior diagnostic performance as a non-invasive biomarker for early colorectal cancer detection. The systematic review and meta-analysis suggest its potential utility, though further large-scale prospective studies are needed to standardize testing protocols and confirm clinical applicability.

IHC

Integration of deep learning and radiomic features from multiplex immunohistochemistry images for reproducible Multi-Outcome prediction in a Multi-Center study of colorectal cancer

Fused multiplex immunohistochemistry (mIHC)-derived radiomic and deep learning features provide accurate and interpretable predictions for multiple colorectal cancer outcomes. The study supports the integration of these features into precision oncology workflows, enhancing the potential for reproducible multi-outcome predictions across different centers.

Stool DNA

An algorithm-enhanced stool DNA system improves the differential diagnosis of colorectal cancer versus Crohn's disease in high-risk symptomatic patients

The FIT-sDNA-CA system, which integrates genetic, epigenetic, and inflammatory biomarkers, enhances the differential diagnosis of colorectal cancer versus Crohn's disease in high-risk symptomatic patients. This algorithm-enhanced stool DNA testing improves specificity compared to traditional fecal DNA tests, serving as an accurate non-invasive triage tool that aids in early risk stratification and reduces unnecessary endoscopic referrals.

What this means for you

If you have been diagnosed with colorectal cancer, ask your oncologist: *"Has my tumor been tested for all the biomarkers that have an FDA-approved treatment?"* This is especially important if you have been diagnosed with metastatic (Stage IV) colorectal cancer, where biomarker results directly determine which targeted therapies and immunotherapies are available to you. Based on the research in this report, the most actively studied biomarkers right now are **ctDNA, TP53, BRAF, NGS, and MSI/MMR**. Testing should happen at diagnosis — before treatment begins — because results take 1–2 weeks and can shape your first treatment decision.

Generated from structured biomarker research data · 2026-04-20 07:30:02

Appendix B — Colon & Colorectal Cancer: Biomarker Testing Reference

What Is Biomarker Testing?

Biomarker testing — also called molecular profiling or genomic testing — analyzes your tumor's DNA to identify specific mutations. A single next-generation sequencing (NGS) panel can check many gene mutations at once. Results typically take 1–2 weeks and should be obtained at diagnosis, before treatment begins.

Key Biomarkers — Test All mCRC at Diagnosis

RAS (KRAS/NRAS) · BRAF V600E · MSI/MMR status · HER2 · NTRK fusion · TMB

Testing Methods

NGS (next-generation sequencing) — checks many mutations at once from tumor tissue

Liquid biopsy / ctDNA — blood test that detects tumor DNA; useful when tissue is unavailable

IHC (immunohistochemistry) — used to test MMR status and HER2

PCR — used for specific mutation detection (e.g. MSI, KRAS)

Questions to Ask Your Oncologist

"Has my tumor been tested for all the biomarkers that have an FDA-approved treatment?"

"Was my sample taken from the primary tumor or a metastatic site — and does it need to be retested?"

"How long will results take, and will they affect my first treatment decision?"

"Is a liquid biopsy appropriate for my situation?"

Abbreviations

BRAF V600E — a specific gene mutation found in ~8–10% of colorectal cancers

ctDNA — circulating tumor DNA; tumor DNA fragments found in the bloodstream

dMMR — mismatch repair deficient; indicates the tumor may respond to immunotherapy

HER2 — a protein that drives growth in ~2–3% of colorectal cancers

IHC — immunohistochemistry; a lab staining method used to detect proteins in tissue

KRAS G12C — a specific gene mutation found in ~3–4% of colorectal cancers

mCRC — metastatic colorectal cancer (Stage IV)

MSI-H — microsatellite instability-high; indicates the tumor may respond to immunotherapy

NGS — next-generation sequencing; a method that checks many gene mutations at once

NTRK — a rare gene fusion found in <1% of colorectal cancers

RAS — a gene family (KRAS/NRAS) whose mutation status affects treatment selection

TMB — tumor mutational burden; a measure of how many mutations are in a tumor

Key Resources

ClinicalTrials.gov · NCCN Guidelines (nccn.org) · Colorectal Cancer Alliance (ccalliance.org) · Fight Colorectal Cancer (fightcrc.org) · ACS (cancer.org)

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About This Report

This report is generated automatically from publicly available medical databases including PubMed, ASCO Journal of Clinical Oncology, Annals of Oncology, and NCI Cancer.gov. Entries are classified using deterministic rule-based logic and scored by clinical significance. AI-generated summaries are grounded exclusively in the source data provided.

Limitations

This report does not represent a complete picture of all published research. Database coverage, feed availability, and relevance filtering affect what appears. Research findings may not yet reflect current clinical guidelines. Always verify information at the original source before making clinical decisions.

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